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ELECTRICAL SCIENCE

Syllabus

1 → DC Circuits

2 → AC Circuits

3 → DC Machines

4 → Transformers

TOPICS



- Circuit Components
- ② → Sources
- Current DIVIDER RULE
- ② → Star & Delta Connection/Transformation
- ② → THEVENIN'S Theorem
- ② → NORTAN'S Theorem
- Maximum Power Theorem
- ② → Superposition Theorem
- ② → Nodal & Mesh analysis

Components

• Active elements

Elements having their own energy or those elements which supply energy to the network

eg - Battery ; Solar cells etc

• Passive Elements

Elements which don't have their own energy source or which don't provide energy to the network

eg - Resistors, Inductors, capacitors, etc

• Linear Elements

Those elements which follow Ohm's law.

I v/s V graph is a straight line

eg - Resistors, Capacitors, Inductors, etc

• Non-Linear Elements

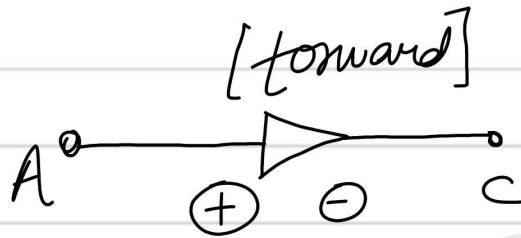
Those elements which don't follow Ohm's law
 I v/s V graph is not a straight line

eg - Transformers, Transistors, etc

• Unilateral Element

Those elements in which current flow in a single direction

eg - diode



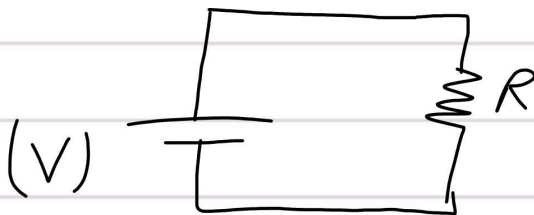
• Bilateral Element

Those elements in which current flow in both direction

eg - Resistor, Transmission line.

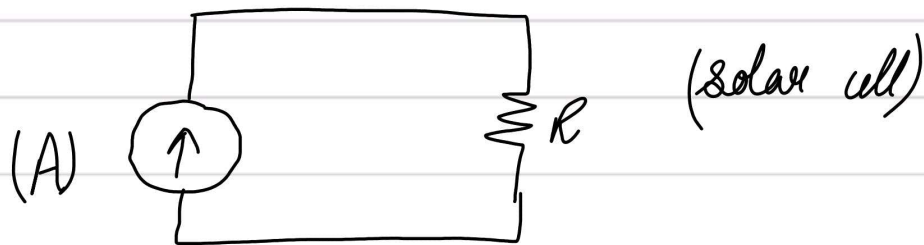
SOURCE

(1) Voltage Source



(Cells, Batteries)

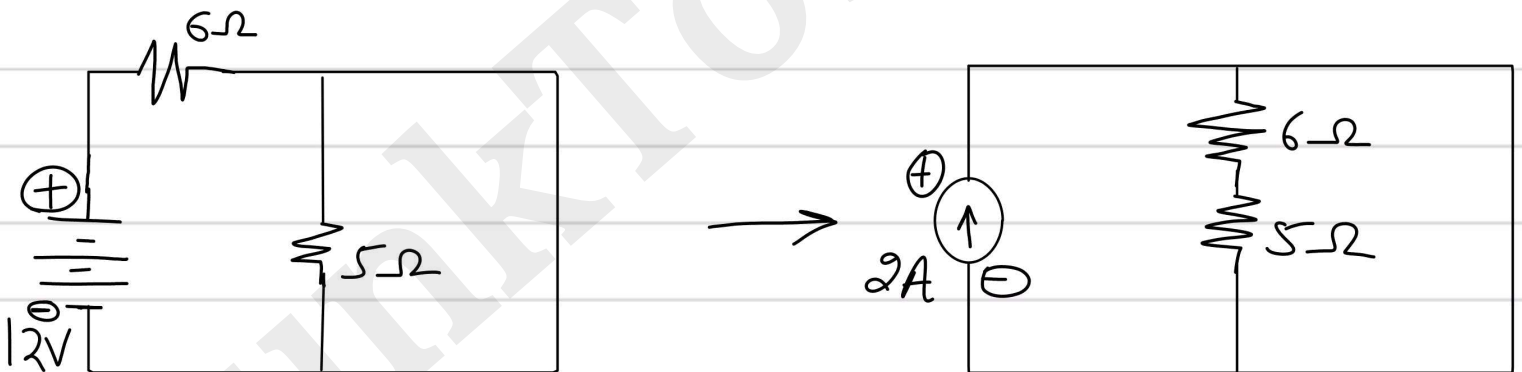
(2) Current Source



• Conversion between them

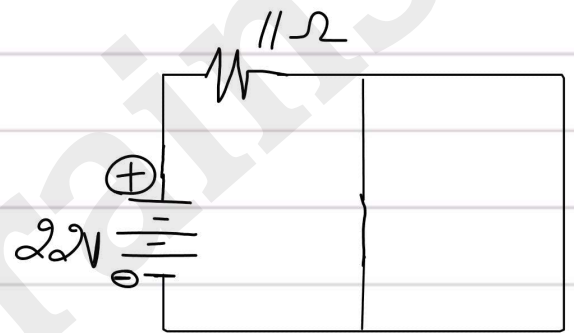
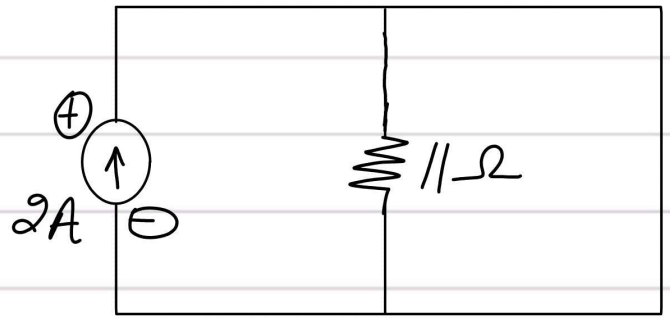
(1) Voltage $\longrightarrow$ Current

- To convert voltage source V in series with resistor R
- Into current source $I = V/R$ in // with the same resistor

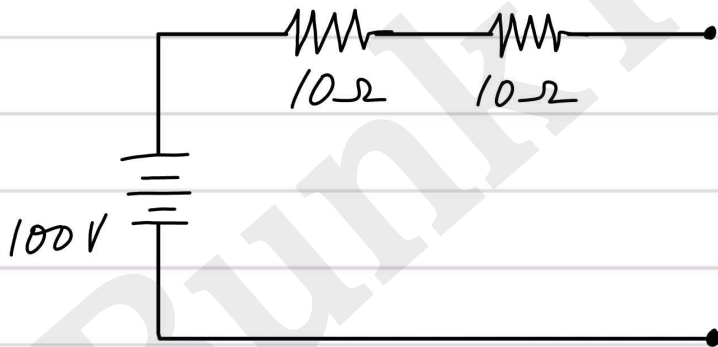


2) Current $\longrightarrow$ Voltage

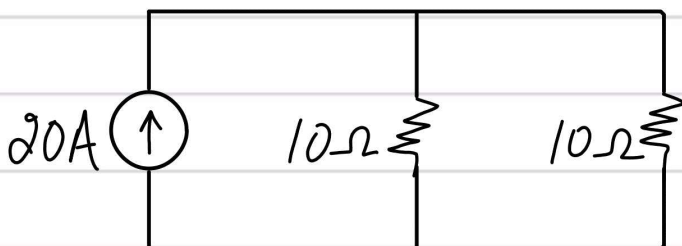
- To convert current source in // with resistor R
- Into voltage source $V = I \times R$ in series with same resistor



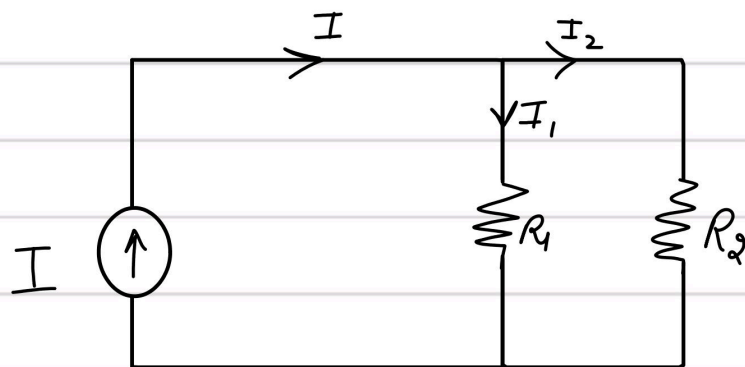
Ques Convert given voltage source into current source, for the given circuit



Ques Convert given current source into voltage source for the given circuit



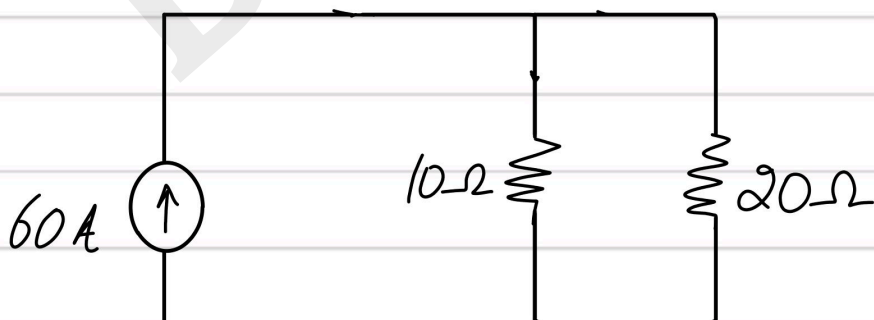
CURRENT DIVIDER RULE



$$I_1 = I \times \frac{R_2}{R_2 + R_1}$$

$$I_2 = I \times \frac{R_1}{R_2 + R_1}$$

Ques find the current across 10Ω and 20Ω resistance



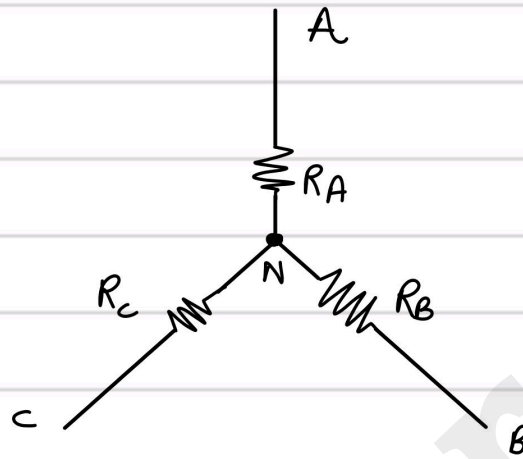
Ans $\rightarrow$

$10\Omega \rightarrow 40A$

$20\Omega \rightarrow 20A$

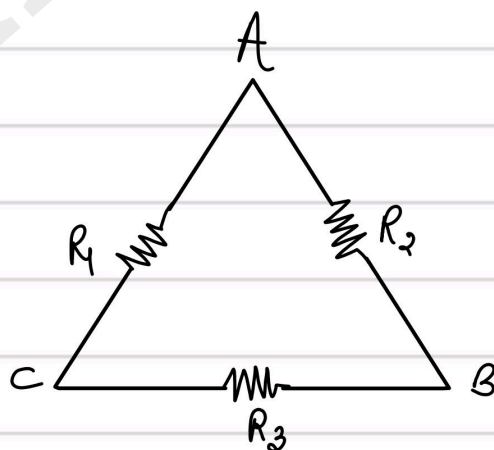
→ STAR CONNECTION

When 3 resistors are connected at a common point (node), that connection is known as star connection

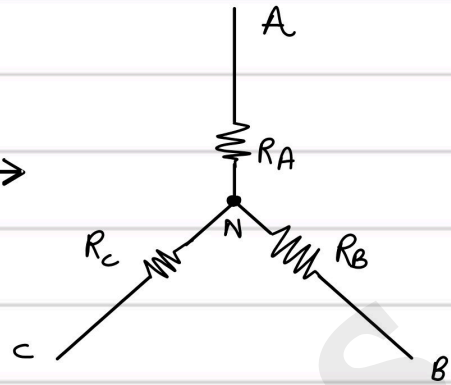
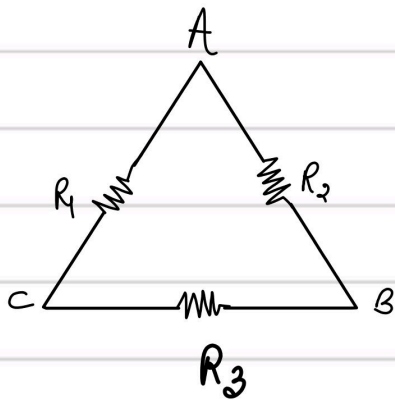


→ DELTA CONNECTION

When 3 resistors are connected such that they make a triangle or mesh, then this type of connection is known as Delta Connection (Δ)



→ DELTA TO STAR ($\Delta \rightarrow Y$)

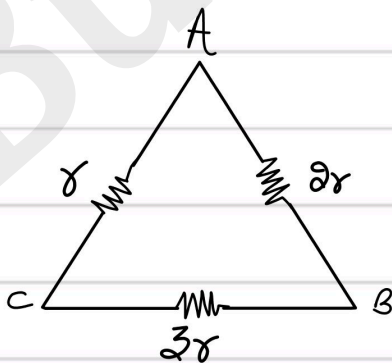


$$R_A = \frac{R_1 R_2}{R_1 + R_2 + R_3}$$

$$R_B = \frac{R_2 R_3}{R_1 + R_2 + R_3}$$

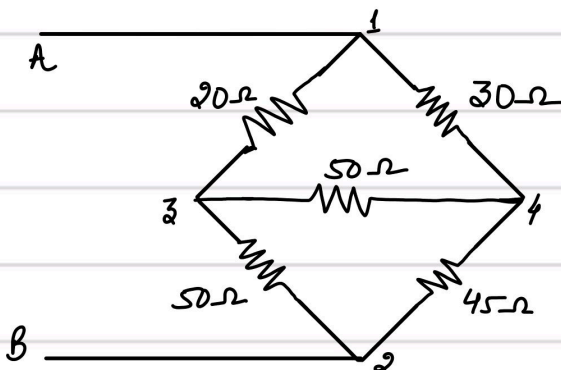
$$R_C = \frac{R_3 R_1}{R_1 + R_2 + R_3}$$

Ques



find the star connection for this delta connection

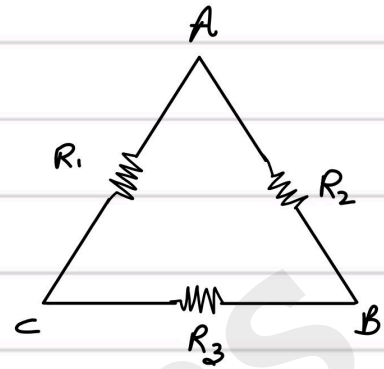
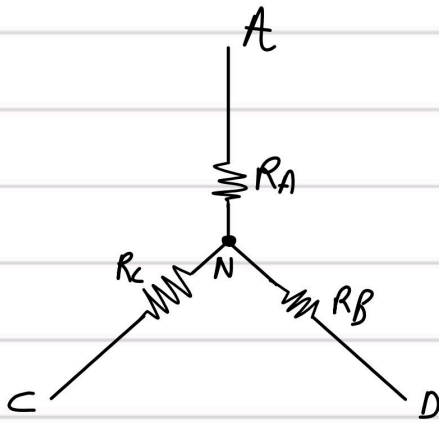
Ques



Determine eq resistance b/w A & B by using star & Delta transformation

Ans $R_{AB} = 36\Omega$

→ STAR TO DELTA (Y → Δ)

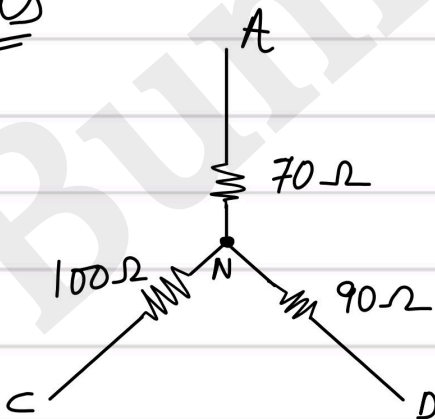


$$R_1 = R_A + R_C + \frac{R_A R_C}{R_B}$$

$$R_2 = R_A + R_B + \frac{R_A R_B}{R_C}$$

$$R_3 = R_B + R_C + \frac{R_B R_C}{R_A}$$

Ques



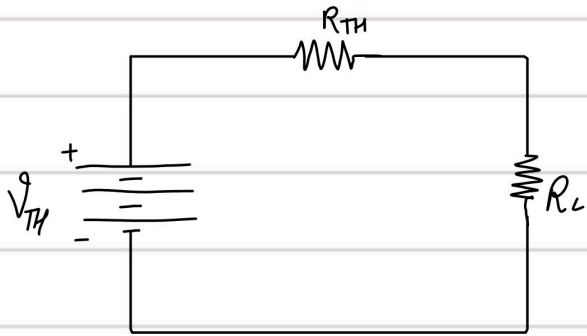
Convert this star connection into delta connection

NOTE → (1) Replace voltage source with straight wire for short circuit.

 for Thevenin &

(2) Replace current source with open wire for open wire

→ THEVENIN'S THEOREM



→ In a linear bilateral network, consisting of a voltage source and resistance R , can be converted into a simplified circuit where voltage V_{TH} will be in series with resistance R_{TH}

$$I_L = \frac{V_{TH}}{R_{TH} + R_L} \rightarrow \text{always given}$$

$V_{TH} \rightarrow$ Thevenin's Voltage
 $R_{TH} \rightarrow$ Thevenin's Resistance
 $R_L \rightarrow$ Load Resistance
 $I_L \rightarrow$ Load Current

• Procedure for R_{TH}

(1) Remove load resistance and replace it with open terminals



(2) Do voltage source $\rightarrow$ short circuit & current source $\rightarrow$ open circuit
and calculate R_{TH} .

• Procedure for V_{TH}

- (1) Remove load resistance
- (2) Convert all current sources into voltage sources.
- (3) Apply KVL across AB terminal

NOTE

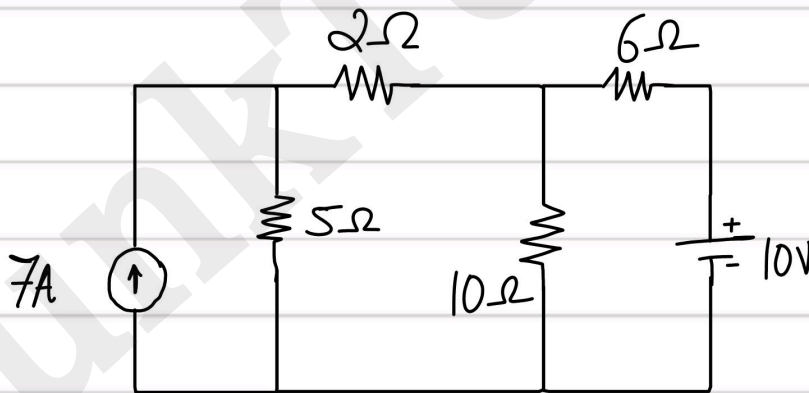
$$\begin{array}{c} \xrightarrow{\quad} \\ \bullet - | + \bullet \\ \xrightarrow{\quad} \end{array} = +ve$$

$$\begin{array}{c} \xrightarrow{\quad} \\ \bullet + | - \bullet \\ \xrightarrow{\quad} \end{array} = -ve$$

$$\begin{array}{c} \xrightarrow{\quad} \\ \text{---} \text{---} \text{---} \text{---} \text{---} \\ \text{IR} \end{array} = -ve$$

$$\begin{array}{c} \xleftarrow{\quad} \\ \text{---} \text{---} \text{---} \text{---} \text{---} \\ \text{IR} \end{array} = +ve$$

Ques By using Thevenin Theorem determine current across 2Ω resistor for the given circuit



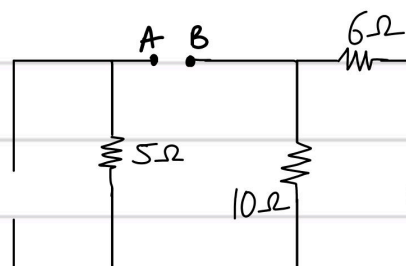
By Thevenin Theorem,

$$I_L = \frac{V_{TH}}{R_{TH} + R_L}$$

given that,

$$R_L = 2\Omega$$

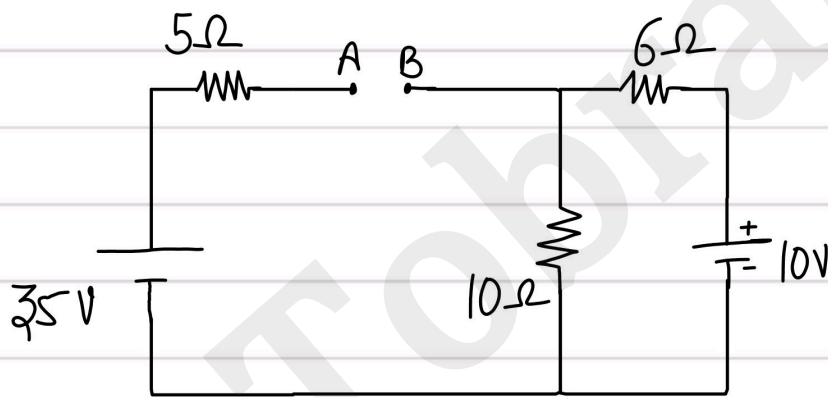
$$R_{TH} = (6 \parallel 10) + 5$$



$$= \frac{6 \times 10}{6+10} + 5 = 3.75 + 5 = \underline{\underline{8.75}} \rightarrow R_{TH}$$

now, for V_{TH} , converting current source into voltage source as current and 5Ω resistance are in parallel with each other

$$\begin{aligned} V &= IR \\ &= 7 \times 5 \\ &= 35V \end{aligned}$$

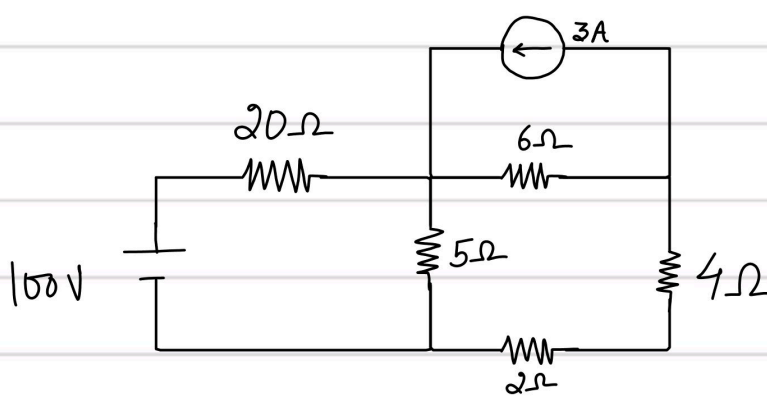


$$\begin{aligned} V_{TH} &= -10I + 35 \\ &= -10 \times \frac{10}{6+10} + 35 \end{aligned}$$

$$= \frac{-100}{16} + 35 = \underline{\underline{28.75 V}}$$

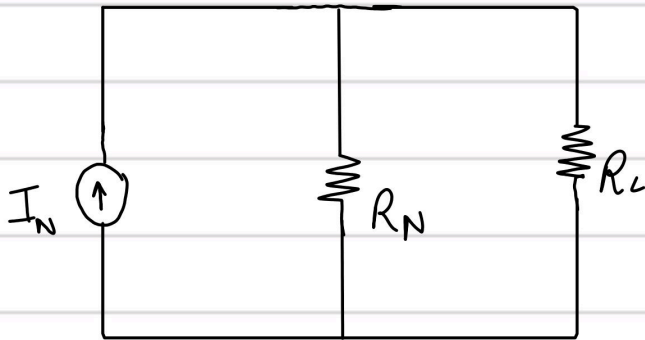
$$I_L = \frac{V_{TH}}{R_{TH} + R_L} = \frac{28.75}{8.75 + 2} = \frac{28.75}{10.75} = 2.67A$$

Ques find the current across 4Ω resistance for the given circuit by using Thevenin Theorem



Ans $\rightarrow I_L = 0.125A$

$\rightarrow$ NORTON'S Theorem



$\rightarrow$ Any linear Bilateral network can be replaced by an equivalent circuit consisting of a current source in // with a resistor.

$$I = \frac{I_N \times R_N}{R_N + R_L}$$

$$\circ R_N = R_{TH} \quad \& \quad I_N = \frac{V_{TH}}{R_{TH}}$$

- ★ Voltage $\rightarrow$ Current Norton's theorem
- ★ Current $\rightarrow$ Voltage Thevenin's theorem

→ MAXIMUM POWER THEOREM

When load resistance (R_L) is equal to internal resistance (R_o) [Thevenin's Resistance] then the maximum power will be transferred in a linear bilateral circuit.

$$R_L = R_o = R_{TH}$$

↳ Thevenin's Resistance

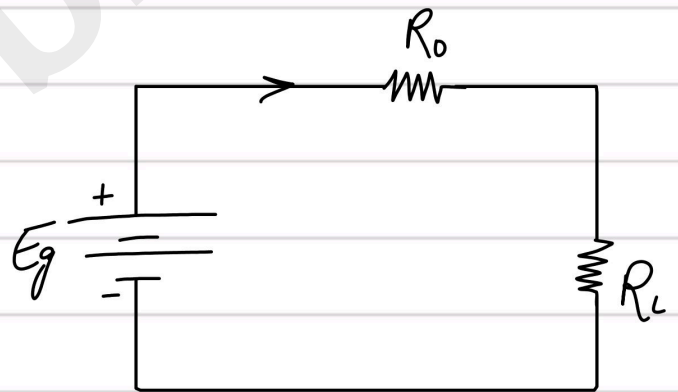
PROOF

let

E_g = generated voltage

R_o = Internal resistance

R_L = Load resistance



Power across load resistance is given by,

$$P = I^2 R_L \quad \text{--- (1)}$$

If the value of current is I , then the value of current

$$I = \frac{E_g}{R_o + R_L} \quad \text{--- (2)}$$

putting the value of I in eq ①

$$P = \frac{(E_g)^2}{(R_0 + R_L)^2} R_L \quad \text{--- (3)}$$

diff. eq ② w.r.t R_L

$$\frac{dP}{dR_L} = \frac{(R_0 + R_L)^2 (E_g)^2 - (E_g)^2 R_L 2(R_0 + R_L)}{(R_0 + R_L)^4}$$

$$\frac{dP}{dR_L} = \frac{(E_g)(R_0 + R_L) \{ (R_0 + R_L) - 2R_L \}}{(R_0 + R_L)^4}$$

$$\frac{dP}{dR_L} = \frac{(E_g)^2 [R_0 - R_L]}{(R_0 + R_L)^3}$$

for maximum power transfer
, $\frac{dP}{dR_L} = 0$

$$\rightarrow \frac{(E_g)^2 (R_0 - R_L)}{(R_0 + R_L)^3} = 0$$

$$\rightarrow (E_g)^2 [R_0 - R_L] = 0$$

$(E_g)^2 \neq 0$ Therefore,

$$\begin{aligned} R_0 - R_L &= 0 \\ R_0 &= R_L \end{aligned}$$

This is the required expression for maximum power transferred

→ SUPERPOSITION THEOREM

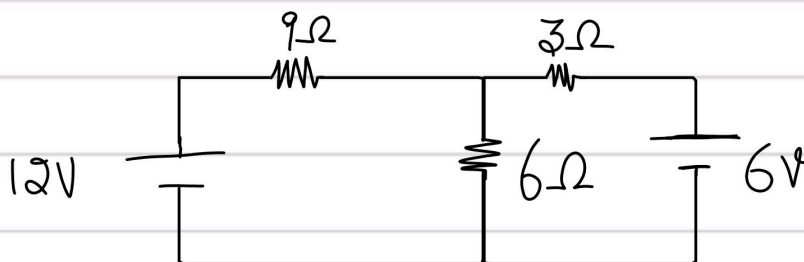
★ Only applicable for linear element.

★ Atleast we have two or more than two sources available in the circuit. Sources maybe voltage sources, current sources and both combination of voltage source and current source.

→ STATEMENT

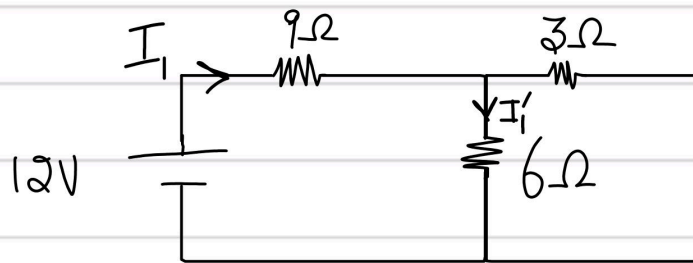
With the help of superposition theorem we determine current in particular branch. This current is algebraic sum of all the currents flowing through that branch, due to all the sources available in the circuit.

Ques Determine the current across 6Ω resistor



Ans Case I

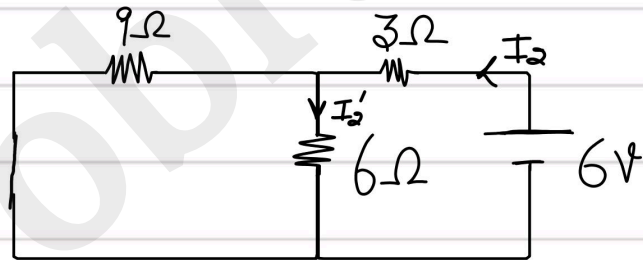
$6V \rightarrow$ short



$$I' = I_1 \times \frac{3}{3+6} = \frac{12}{(3 \parallel 6) + 9} \times \frac{3}{9}$$

$$= \frac{12}{2+9} \times \frac{3}{9} = \frac{4}{11} \times \frac{2}{3} = \frac{4}{11}$$

Case II 12V → Short



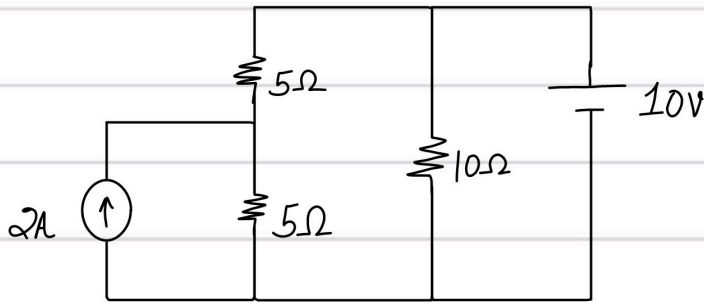
$$I' = I_2 \times \frac{9}{9+6} = \frac{6}{(9 \parallel 6) + 3} \times \frac{9}{15}$$

$$= \frac{6}{11}$$

total current across 6Ω resistance

$$\begin{aligned} I &= I_1' + I_2' \\ &= \frac{4}{11} + \frac{6}{11} = \frac{10}{11} \text{ A} \end{aligned}$$

Ques find current across all resistors



→ SOME BASIC TERMS

• NODES :

It is a point where two or more than two branches meet, that pt in a circuit is called node.

• JUNCTION :

When 3 or more than 3 branches are meeting at a point, that pt is known as junction.

• BRANCH :

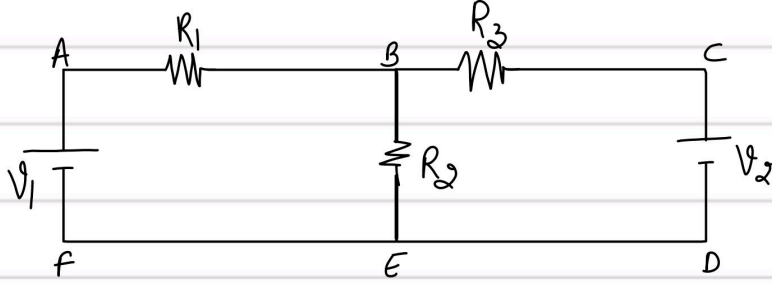
Representation of any electrical element by line segment is known as Branch.

• MESH :

The smallest closed connection in which there is no any other closed connection exists.

• LOOP :

Loop is a largest closed connection, in which other closed connection maybe exist.



- Loop : ABCDEFA
- Mesh : ABEFA
BCDEB
- Node : A, C, D, F
- Junction : B, E

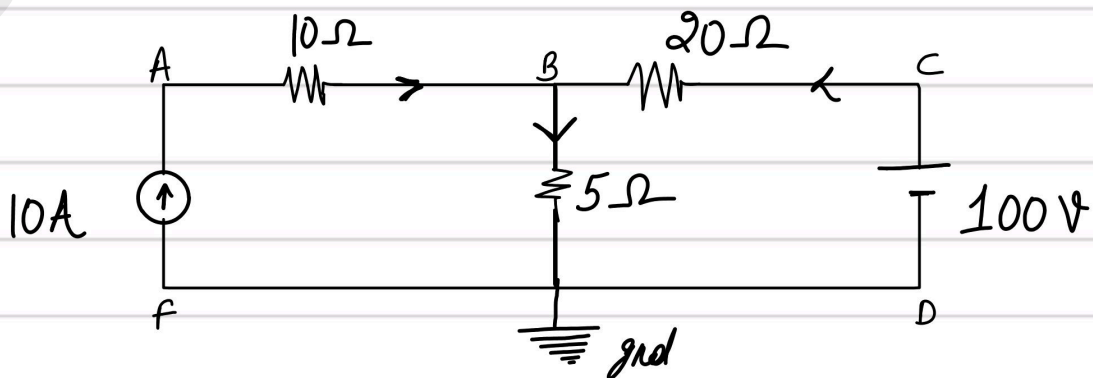
→ KCL

It states that the algebraic sum of incoming current and outgoing current is equal to 0, at node or junction.

→ KVL

It states that the algebraic sum of all the voltage source & voltage drop is 0, in a mesh or loop.

Ques Determine current in all branches using nodal analysis (KCL) in the given circuit.



Independent & Dependent Sources

1) Independent Source

- The value of the source does not depend on anything else.
- Provides a fixed voltage or current.

2) Dependent (Controlled) Source

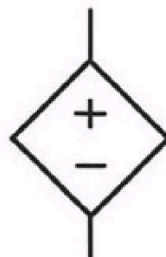
- The value of the source depends on another voltage or current in the circuit.



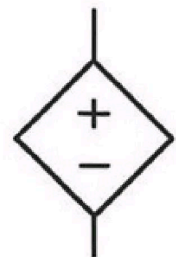
Current-Controlled
Current Source
 $I = 3I_1$



Current-Controlled
Voltage Source
 $V = 3I_1$



Voltage-Controlled
Voltage Source
 $V = 3V_1$



Voltage-Controlled
Current Source
 $I = 3V_1$

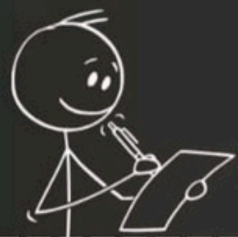


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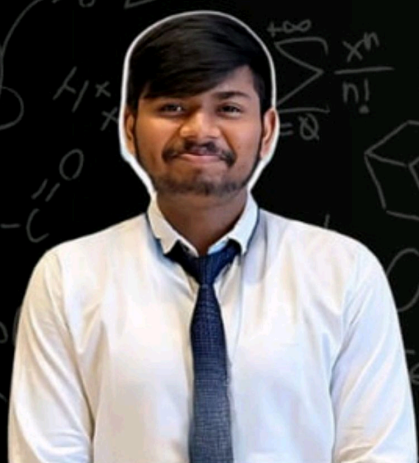


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